INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	Cologix Jacksonville JAX1, FL -- station KSSI, America/New_York
Building	OSM 497151852 Cologix Jacksonville JAX1
Facade gap	not applicable -- one building, no second facade
Weather record	43,390 real hourly records from KSSI
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-01-01 (recirc_edge)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	gate off

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 4 of 23 hours with 2 mode changes. 4 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	4 of 23 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
Incumbent	4 of 23 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 4 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	17.500	18.0	17.220	0.728
01	mechanical	yes	switch budget	17.775	18.0	17.780	0.716
03	mechanical	yes	switch budget	17.775	18.0	17.780	0.704
04	mechanical	yes	switch budget	17.775	18.0	17.780	0.698
05	mechanical	no	dry-bulb	18.060	18.0	17.780	0.681
06	mechanical	no	dry-bulb	18.060	18.0	17.780	0.601
07	mechanical	no	dry-bulb	18.335	18.0	17.220	0.726
08	mechanical	no	dry-bulb	20.000	18.0	17.780	1.223
09	mechanical	no	dry-bulb	21.280	18.0	17.780	1.692

10	mechanical	no	dry-bulb	22.220	18.0	19.440	1.789
11	mechanical	no	dry-bulb	21.945	18.0	20.000	1.306
12	mechanical	no	dry-bulb	21.390	18.0	20.000	1.239
13	mechanical	no	dry-bulb	21.665	18.0	20.000	1.193
14	mechanical	no	dry-bulb	20.555	18.0	18.330	1.087
15	mechanical	no	dry-bulb	19.890	18.0	17.780	1.102
16	mechanical	no	dry-bulb	19.390	18.0	17.780	0.970
17	mechanical	no	dry-bulb	18.055	18.0	17.780	0.839
18	FREE	yes	-	17.500	18.0	17.220	1.052
19	FREE	yes	-	17.500	18.0	17.220	1.118
20	FREE	yes	-	17.780	18.0	17.780	0.939
21	FREE	yes	-	17.775	18.0	18.330	0.712
22	mechanical	no	dry-bulb	18.050	18.0	18.330	0.795
23	mechanical	no	dry-bulb	18.060	18.0	17.780	0.721

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.500 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.775 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.775 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.775 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.060 C against a 18.0 C limit, so it fails by 0.060 C. It would take a limit 0.060 C higher, or a bound 0.060 C tighter, to change this hour.

06:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.060 C against a 18.0 C limit, so it fails by 0.060 C. It would take a limit 0.060 C higher, or a bound 0.060 C tighter, to change this hour.

07:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.335 C against a 18.0 C limit, so it fails by 0.335 C. It would take a limit 0.335 C higher, or a bound 0.335 C tighter, to change this hour.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.000 C against a 18.0 C limit, so it fails by 2.000 C. It would take a limit 2.000 C higher, or a bound 2.000 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.280 C against a 18.0 C limit, so it fails by 3.280 C. It would take a limit 3.280 C higher, or a bound 3.280 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.220 C against a 18.0 C limit, so it fails by 4.220 C. It would take a limit 4.220 C higher, or a bound 4.220 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.945 C against a 18.0 C limit, so it fails by 3.945 C. It would take a limit 3.945 C higher, or a bound 3.945 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.390 C against a 18.0 C limit, so it fails by 3.390 C. It would take a limit 3.390 C higher, or a bound 3.390 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.665 C against a 18.0 C limit, so it fails by 3.665 C. It would take a limit 3.665 C higher, or a bound 3.665 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.555 C against a 18.0 C limit, so it fails by 2.555 C. It would take a limit 2.555 C higher, or a bound 2.555 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.890 C against a 18.0 C limit, so it fails by 1.890 C. It would take a limit 1.890 C higher, or a bound 1.890 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.390 C against a 18.0 C limit, so it fails by 1.390 C. It would take a limit 1.390 C higher, or a bound 1.390 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.055 C against a 18.0 C limit, so it fails by 0.055 C. It would take a limit 0.055 C higher, or a bound 0.055 C tighter, to change this hour.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.500 C, which is 0.500 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.051 C, and the margin is measured, not chosen: 1.051 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.500 C, which is 0.500 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.118 C, and the margin is measured, not chosen: 1.118 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.780 C, which is 0.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.939 C, and the margin is measured, not chosen: 0.939 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.775 C, which is 0.225 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.712 C, and the margin is measured, not chosen: 0.712 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by. *** THIS WAS WRONG: the intake actually reached 18.330 C, above the limit. The bound failed here, and it is counted as a breach rather than explained away. ***

22:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.050 C against a 18.0 C limit, so it fails by 0.050 C. It would take a limit 0.050 C higher, or a bound 0.050 C tighter, to change this hour.

23:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.060 C against a 18.0 C limit, so it fails by 0.060 C. It would take a limit 0.060 C higher, or a bound 0.060 C tighter, to change this hour.

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